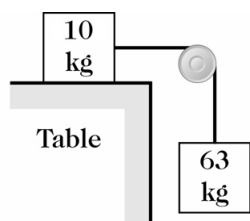


- 1) A ball is thrown downward in the absence of air resistance. After it has been released, which statement(s) concerning its acceleration is correct? (There could be more than one correct choice.)
- A) Its acceleration is constantly increasing.
 - B) Its acceleration is greater than g .
 - C) Its acceleration is constantly decreasing.
 - D) Its acceleration is constant.
 - E) Its acceleration is zero.
- 2) A 10-kg rock and a 20-kg rock are thrown upward with the same initial speed v_0 and experience no significant air resistance. If the 10-kg rock reaches a maximum height h , what maximum height will the 20-kg ball reach?
- A) h B) $h/4$ C) $2h$ D) $h/2$ E) $4h$
- 3) The velocity $v(t)$ of a particle as a function of time is given by $v(t) = (2.3 \text{ m/s}) + (4.1 \text{ m/s}^2)t - (6.2 \text{ m/s}^3)t^2$. What is the average acceleration of the particle between $t = 1.0 \text{ s}$ and $t = 2.0 \text{ s}$?
- A) 13 m/s^2 B) 0 m/s^2 C) -15 m/s^2 D) 15 m/s^2 E) -13 m/s^2
- 4) The x component of vector $\vec{A}$ is 8.7 units, and its y component is -6.5 units. The magnitude of $\vec{A}$ is closest to
- A) 8.9 units. B) 11 units. C) 12 units. D) 9.9 units. E) 7.9 units.
- 5) When Jeff ran up a hill at 7.0 m/s, the horizontal component of his velocity vector was 5.1 m/s. What was the vertical component of Jeff's velocity?
- A) 4.3 m/s B) 3.4 m/s C) 4.8 m/s D) 3.8 m/s
- 6) The magnitude of $\vec{A}$ is 5.5 m, and this vector lies in the second quadrant and makes an angle of 34° with the $+y$ -axis. The components of $\vec{A}$ are closest to
- A) $A_x = -4.6 \text{ m}$, $A_y = 3.1 \text{ m}$.
 - B) $A_x = 4.6 \text{ m}$, $A_y = -3.1 \text{ m}$.
 - C) $A_x = -4.6 \text{ m}$, $A_y = -3.1 \text{ m}$.
 - D) $A_x = 3.1 \text{ m}$, $A_y = -4.6 \text{ m}$.
 - E) $A_x = -3.1 \text{ m}$, $A_y = 4.6 \text{ m}$.
- 7) For general projectile motion with no air resistance, the horizontal component of a projectile's acceleration
- A) first decreases and then increases.
 - B) continuously increases.
 - C) is always zero.
 - D) remains a non-zero constant.
 - E) continuously decreases.

- 8) A stone is thrown horizontally with an initial speed of 10 m/s from the edge of a cliff. A stopwatch measures the stone's trajectory time from the top of the cliff to the bottom to be 4.3 s. What is the height of the cliff if air resistance is negligibly small?
- A) 91 m B) 22 m C) 77 m D) 43 m
- 9) A cat leaps to try to catch a bird. If the cat's jump was at 60° off the ground and its initial velocity was 2.74 m/s, what is the highest point of its trajectory, neglecting air resistance?
- A) 0.19 m B) 10.96 m C) 0.58 m D) 0.29 m
- 10) A 20-ton truck collides with a 1500-lb car. Which of the following statements must be true?
- A) During the collision, the force on the truck is smaller than the force on the car.
 B) The truck did not slow down during the collision, but the car did.
 C) During the collision, the force on the truck is greater than the force on the car.
 D) The car did not slow down during the collision, but the truck did.
 E) During the collision, the force on the truck is equal to the force on the car.
- 11) The acceleration due to gravity is lower on the Moon than on Earth. Which one of the following statements is true about the mass and weight of an astronaut on the Moon's surface, compared to Earth?
- A) Both mass and weight are the same. B) Both mass and weight are less.
 C) Mass is less, weight is the same. D) Mass is the same, weight is less.
- 12) A certain aircraft has a mass of 300,000 kg. At a certain instant during its landing, its speed is 27.0 m/s. If the braking force is a constant 445,000 N, what is the speed of the airplane 10.0 s later?
- A) 10.0 m/s B) 18.0 m/s C) 20.0 m/s D) 12.2 m/s E) 14.0 m/s
- 13) A 45.0-kg person steps on a scale in an elevator. The scale reads 460 N. What is the magnitude of the acceleration of the elevator?
- A) 0.206 m/s^2 B) 46.9 m/s^2 C) 4.91 m/s^2 D) 9.81 m/s^2 E) 0.422 m/s^2
- 14) A race car traveling at a constant speed of 50 m/s drives around a circular track that is 500 m in diameter. What is the magnitude of the acceleration of the car?
- A) 10 m/s^2 B) 1.0 m/s^2 C) 2.2 m/s^2 D) 0 m/s^2 E) 0.63 m/s^2
- 15) A car moving at a steady 10 m/s on a level highway encounters a bump that has a circular cross-section with a radius of 30 m. The car maintains its speed over the bump. What is the normal force exerted by the seat of the car on a 60.0-kg passenger when the car is at the top of the bump?
- A) 790 N B) 390 N C) 590 N D) 490 N E) 200 N

- 16) As shown in the figure, a 10-kg block on a perfectly smooth horizontal table is connected by a horizontal string to a 63-kg block that is hanging over the edge of the table. What is the magnitude of the acceleration of the 10-kg block when the other block is gently released?



- A) 7.5 m/s^2 B) 8.1 m/s^2 C) 9.0 m/s^2 D) 8.5 m/s^2
- 17) A 50-kg box is being pushed along a horizontal surface. The coefficient of static friction between the box and the ground is 0.35. What horizontal force must be exerted on the box for it to accelerate at 1.2 m/s^2 ?
- A) 120 N B) 490 N C) 170 N D) 60 N E) 230 N
- 18) A 50.0-kg block is being pulled up a 16.0° slope by a force of 250 N that is parallel to the slope. The coefficient of kinetic friction between the block and the slope is 0.200. What is the acceleration of the block?
- A) 0.528 m/s^2 B) 0.260 m/s^2 C) 0.415 m/s^2 D) 0.983 m/s^2 E) 0.158 m/s^2
- 19) A 0.034-g raindrop, starting from rest, falls down from sky. The air resistance is $(3.6 \times 10^{-5} \text{ N} \cdot \text{s/m})v$, where v is raindrop's speed. What is the raindrop's terminal (or maximum) speed?
- A) 0.034 m/s B) 4.9 m/s C) 0.52 m/s D) 9.3 m/s E) 3.6 m/s
- 20) Sally is driving along a straight highway in her 1965 Mustang. At $t = 0$, she is at rest at the origin. Her x -acceleration as a function of time is $a_x = (1.50 \text{ m/s}^3)t - (0.120 \text{ m/s}^4)t^2$. Find the maximum velocity she attains.
- A) 39.1 m/s B) 1.50 m/s C) 0.120 m/s D) 25.6 m/s E) 343 m/s